

differences are in A.P

$$S = 1 + 3 + 6 + 10 + 15 + 21 + \dots \quad (n \text{ terms})$$

Soln

$L_2 \rightarrow$	1	3	6	10	15	21	...
L_1		2	3	4	5	6	
L_0			1	1	1	1	

$$t_n = an^2 + bn + c$$

$$t_1 = a + b + c = 1 \quad -3a - b = -2 \quad 2a = 1$$

$$t_2 = 4a + 2b + c = 3 \quad -5a - b = -3$$

$$t_3 = 9a + 3b + c = 6$$

$$a = \frac{1}{2}, b = \frac{1}{2}, c = 0$$

$$t_n = \frac{1}{2}n^2 + \frac{1}{2}n$$

$$S = \sum_{n=1}^n t_n$$

$$= \sum_{n=1}^n \left(\frac{1}{2}n^2 + \frac{1}{2}n \right)$$

$$= \frac{1}{2} \sum_{n=1}^n n^2 + \frac{1}{2} \sum_{n=1}^n n$$

$$= \frac{1}{2} \times \frac{n(n+1)(2n+1)}{6} + \frac{1}{2} \times \frac{n(n+1)}{2}$$

$$= \frac{n(n+1)}{4} \left[\frac{2n+1}{3} + 1 \right]$$

$$= \frac{n(n+1)}{4} \times \frac{2n+4}{3}$$

$$= \frac{n(n+1)(n+2)}{6}$$

Find sum

$$S = 3 + 7 + 13 + 21 + 31 + \dots$$

(upto n terms)

$$S = 3 + 7 + 13 + 21 + 31 + \dots$$

Soln

$$\begin{array}{l} L_2 \\ L_1 \\ L_0 \end{array} \quad \begin{array}{ccccccc} 3, & 7, & 13, & 21, & 31, & \dots \\ & 4 & 6 & 8 & 10 & \\ & & 2 & 2 & 2 & \end{array}$$

$$t_n = an^2 + bn + c$$

$$t_1 = a + b + c = 3$$

$$-3a - b = -4$$

$$t_2 = 4a + 2b + c = 7$$

$$-5a - b = -6$$

$$t_3 = 9a + 3b + c = 13$$

$$a=1, b=1, c=1$$

$$t_n = n^2 + n + 1$$

$$S = \sum_{n=1}^n t_n$$

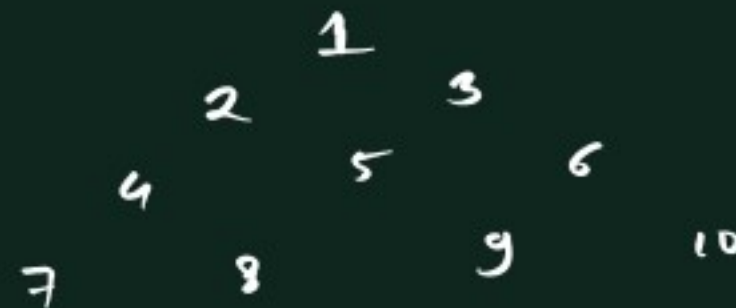
$$= \sum_{n=1}^n (n^2 + n + 1)$$

$$= \sum_{n=1}^n n^2 + \sum_{n=1}^n n + \sum_{n=1}^n 1$$

$$= \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} + n$$

$$= n \left[\frac{(n+1)(2n+1)}{6} + \frac{n+1}{2} + 1 \right]$$

- Q
- a) find sum of all the numbers till nth row
- b) find sum of elements of nth row only



Soln

m1 $S = 1 + 5 + 15 + 34 + 65 + \dots$ (nth row)

$L_3: 1, 5, 15, 34, 65$

$L_2: 4, 10, 19, 31$

$L_1: 6, 9, 12$

$L_0: 3, 3$

$t_n = an^3 + bn^2 + cn + d$

m2

a) last term of n th row = total numbers till n th row
 $= 1 + 2 + 3 + \dots + n$
 $= \frac{n}{2} [1 + n]$
 $= \frac{n(n+1)}{2}$



$$S = 1 + 2 + 3 + \dots + \frac{n(n+1)}{2}$$

$$= \frac{\frac{n(n+1)}{2}}{2} \left[1 + \frac{n(n+1)}{2} \right]$$

$$= \frac{n(n+1)}{4} \left[\frac{n^2 + n + 2}{2} \right] = \frac{n(n+1)(n^2 + n + 2)}{8}$$

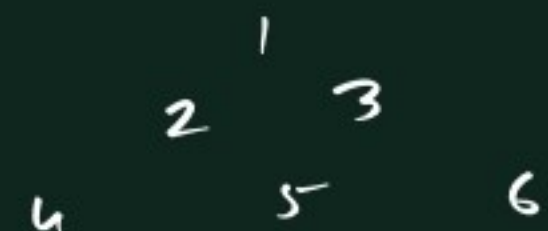


b)

last number of $(n-1)$ th row = $1 + 2 + 3 + \dots + (n-1)$

$$= \frac{n-1}{2} [1 + n-1]$$

$$= \frac{n(n-1)}{2}$$



$(n-1)$ th row
 n th row $\rightarrow$

$\frac{n(n-1)}{2} + 1$

sum of elements in n th row

$$= \left(\frac{n(n-1)}{2} + 1 \right) + \dots + \left(\frac{n(n+1)}{2} \right)$$

$$= \frac{n}{2} \left[\frac{n(n-1)}{2} + 1 + \frac{n(n+1)}{2} \right]$$

$$= \frac{n}{2} \left[\frac{n^2 - n + 2 + n^2 + n}{2} \right]$$

$$= \frac{n}{2} (n^2 + 1)$$

$\frac{n(n-1)}{2}$

$\frac{n(n+1)}{2}$

telescopic

$$\begin{aligned} \underline{\underline{Q}} \quad S &= \frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \frac{1}{7 \cdot 9} + \dots \\ S &= \frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \frac{1}{7 \cdot 9} + \dots + \frac{1}{(2n-1)(2n+1)} \\ &= \frac{1}{2} \left[\frac{2}{1 \cdot 3} + \frac{2}{3 \cdot 5} + \frac{2}{5 \cdot 7} + \dots + \frac{2}{(2n-1)(2n+1)} \right] \\ &= \frac{1}{2} \left[\frac{3-1}{1 \cdot 3} + \frac{5-3}{3 \cdot 5} + \frac{7-5}{5 \cdot 7} + \dots + \frac{(2n+1)-(2n-1)}{(2n-1)(2n+1)} \right] \\ &= \frac{1}{2} \left[\left(\frac{1}{1} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{5} \right) + \left(\frac{1}{5} - \frac{1}{7} \right) + \dots + \left(\frac{1}{2n-1} - \frac{1}{2n+1} \right) \right] \\ &= \frac{1}{2} \left[1 - \frac{1}{2n+1} \right] \\ &= \frac{1}{2} \times \frac{2n}{2n+1} = \frac{n}{2n+1} \end{aligned}$$

+ (upto n terms)

$$1, 3, 5, 7, \dots$$

$$1 + (n-1)2 = 2n-1$$

$$3, 5, 7, 9, \dots$$

$$3 + (n-1)2 = 2n+1$$

$$T_n = \frac{1}{(2n-1)(2n+1)}$$

find sum

$$S = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{n(n+1)}$$

$$\text{Q} \quad S = \frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{3 \cdot 4 \cdot 5} + \frac{1}{4 \cdot 5 \cdot 6} + \dots$$

a) upto n terms

b) upto infinite terms

$$T_n = \frac{1}{n(n+1)(n+2)}$$

$$= \frac{1}{2} \cdot \frac{2}{n(n+1)(n+2)}$$

$$= \frac{1}{2} \times \left[\frac{(n+2) - n}{n(n+1)(n+2)} \right]$$

$$= \frac{1}{2} \left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right]$$

$$S_n = \frac{1}{2} \left[\frac{1}{1 \cdot 2} - \frac{1}{(n+1)(n+2)} \right]$$

$$T_1 = \frac{1}{2} \left[\frac{1}{1 \cdot 2} - \frac{1}{2 \cdot 3} \right]$$

$$T_2 = \frac{1}{2} \left[\frac{1}{2 \cdot 3} - \frac{1}{3 \cdot 4} \right]$$

$$T_3 = \frac{1}{2} \left[\frac{1}{3 \cdot 4} - \frac{1}{4 \cdot 5} \right]$$

$\vdots$

$$T_n = \frac{1}{2} \left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right]$$

$$S = \frac{1}{2} \left[\frac{1}{1 \cdot 2} - \frac{1}{(n+1)(n+2)} \right]$$

1, 2, 3, 4

$$1 + (n-1) \cdot 1 = n$$

2, 3, 4, 5

$$2 + (n-1) \cdot 1 = n+1$$

3, 4, 5

$$3 + (n-1) \cdot 1 = n+2$$

$$T_n = \frac{1}{n(n+1)(n+2)}$$

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b) upto infinity

$$T_n = \frac{1}{2} \left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right]$$

$$T_1 = \frac{1}{2} \left(\frac{1}{1 \cdot 2} - \cancel{\frac{1}{2 \cdot 3}} \right)$$

$$T_2 = \frac{1}{2} \left(\cancel{\frac{1}{2 \cdot 3}} - \cancel{\frac{1}{3 \cdot 4}} \right)$$

$$T_3 = \frac{1}{2} \left(\cancel{\frac{1}{3 \cdot 4}} - \cancel{\frac{1}{4 \cdot 5}} \right)$$

$$\vdots$$

$$S_{\infty} = \frac{1}{2} \left(\frac{1}{1 \cdot 2} \right)$$
$$= \frac{1}{4}$$

Q

$$S = \frac{1}{1 \cdot 3} + \frac{1}{2 \cdot 4} + \frac{1}{3 \cdot 5} + \frac{1}{4 \cdot 6} + \dots + (\text{upto infinity})$$

$$T_n = \frac{1}{n(n+2)}$$

$$= \frac{1}{2} \frac{2}{n(n+2)}$$

$$= \frac{1}{2} \left[\frac{(n+2) - n}{n(n+2)} \right]$$

$$= \frac{1}{2} \left[\frac{1}{n} - \frac{1}{n+2} \right]$$

$$T_1 = \frac{1}{2} \left(\frac{1}{1} - \frac{1}{3} \right)$$

$$T_2 = \frac{1}{2} \left(\frac{1}{2} - \frac{1}{4} \right)$$

$$T_3 = \frac{1}{2} \left(\frac{1}{3} - \frac{1}{5} \right)$$

$$T_4 = \frac{1}{2} \left(\frac{1}{4} - \frac{1}{6} \right)$$

$$\vdots$$

$$S = \frac{1}{2} \left[\frac{1}{1} + \frac{1}{2} \right]$$

$$= \frac{1}{2} \times \frac{3}{2} = \frac{3}{4}$$

$$1, 2, 3, \dots$$

$$1 + (n-1)1 = n$$

$$3, 4, 5, 6, \dots$$

$$3 + (n-1)1 = n+2$$

Q find s

$$s = \sum_{n=1}^{\infty} \frac{n}{n^4 + n^2 + 1}$$

Soln

$$t_n = \frac{n}{n^4 + n^2 + 1}$$

$$= \frac{1}{2} \times \frac{2n}{(n^2 + n + 1)(n^2 - n + 1)}$$

$$= \frac{1}{2} \times \frac{(n^2 + n + 1) - (n^2 - n + 1)}{(n^2 + n + 1)(n^2 - n + 1)}$$

$$= \frac{1}{2} \left[\frac{1}{n^2 - n + 1} - \frac{1}{n^2 + n + 1} \right]$$

$$T_1 = \frac{1}{2} \left(1 - \frac{1}{3} \right)$$

$$T_2 = \frac{1}{2} \left(\frac{1}{3} - \frac{1}{7} \right)$$

$$T_3 = \frac{1}{2} \left(\frac{1}{7} - \frac{1}{13} \right)$$

$$\vdots$$

$$\vdots$$

$$\vdots$$

$$\vdots$$

$$s = \frac{1}{2} [1]$$

$$= \frac{1}{2}$$

$$\frac{n^4 + n^2 + 1}{n^4 + 2n^2 + 1 - n^2}$$

$$\frac{(n^2 + 1 + n)(n^2 + 1 - n)}{(n^2 + n + 1)(n^2 - n + 1)}$$

Q find S

$$S = \frac{1}{\underline{3^2+1}} + \frac{1}{\underline{4^2+2}} + \frac{1}{\underline{5^2+3}} + \frac{1}{\underline{6^2+4}} + \dots + (\text{upto infinity})$$

$$T_n = \frac{1}{(n+2)^2 + n}$$

$$= \frac{1}{n^2 + 5n + 4}$$

$$= \frac{1}{(n+1)(n+4)}$$

$$= \frac{1}{3} \times \frac{(n+4) - (n+1)}{(n+1)(n+4)}$$

$$= \frac{1}{3} \left[\frac{1}{n+1} - \frac{1}{n+4} \right]$$

$$T_1 = \frac{1}{3} \left(\frac{1}{2} - \frac{1}{5} \right)$$

$$T_2 = \frac{1}{3} \left(\frac{1}{3} - \frac{1}{6} \right)$$

$$T_3 = \frac{1}{3} \left(\frac{1}{4} - \frac{1}{7} \right)$$

$$T_4 = \frac{1}{3} \left(\frac{1}{5} - \frac{1}{8} \right)$$

$$T_5 = \frac{1}{3} \left(\frac{1}{6} - \frac{1}{9} \right)$$

$$T_6 = \frac{1}{3} \left(\frac{1}{7} - \frac{1}{10} \right)$$

$$\begin{aligned} S &= \frac{1}{3} \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4} \right) \\ &= \frac{1}{3} \times \frac{13}{12} = \frac{13}{36} \end{aligned}$$

3, 4, 5 - - -

$$3 + (n-1)1 = n+2$$

1, 2, 3 - - -

$$1 + (n-1)1 = n$$

find s

$$(i) \quad S = \sum_{n=1}^{\infty} \frac{2n+1}{(n^2+n)^2}$$

$$(ii) \quad S = \frac{1}{1^3} + \frac{1+2}{1^3+2^3} + \frac{1+2+3}{1^3+2^3+3^3} + \dots \quad (\text{upto } n \text{ terms})$$

$$S = \sum_{n=1}^{\infty} \frac{2n+1}{(n^2+n)^2}$$

$$t_n = \frac{2n+1}{(n^2+n)^2}$$

$$= \frac{2n+1}{n^4+n^2+2n^3}$$

$$= \frac{2n+1}{n^2(n^2+2n+1)}$$

$$= \frac{(n^2+2n+1) - n^2}{n^2(n^2+2n+1)}$$

$$= \frac{1}{n^2} - \frac{1}{n^2+2n+1}$$

$$= \frac{1}{n^2} - \frac{1}{(n+1)^2}$$

$$T_1 = \frac{1}{1^2} - \cancel{\frac{1}{2^2}}$$

$$T_2 = \cancel{\frac{1}{2^2}} - \cancel{\frac{1}{3^2}}$$

$$T_3 = \cancel{\frac{1}{3^2}} - \cancel{\frac{1}{4^2}}$$

⋮

$$S = 1$$

$$S = \frac{1}{1^3} + \frac{1+2}{1^3+2^3} + \frac{1+2+3}{1^3+2^3+3^3} + \dots + (\text{upto } n \text{ terms})$$

$$t_n = \frac{1+2+3+\dots+n}{1^3+2^3+3^3+\dots+n^3}$$

$$= \frac{\cancel{n(n+1)}}{\cancel{2} \cdot \frac{\cancel{n^2(n+1)^2}}{\cancel{4}}}$$

$$= \frac{2}{n(n+1)}$$

find S

$$S = 1 \cdot 3 \cdot 5 \cdot 7 + 2 \cdot 4 \cdot 6 \cdot 8 + 3 \cdot 5 \cdot 7 \cdot 9 + 4 \cdot 6 \cdot 8 \cdot 10 + \dots \quad (\text{upto } n \text{ terms})$$

$$T_n = \overset{(n-2)}{n} (n+2) (n+4) (n+6)$$

$$= n (n+2) (n+4) (n+6) \left[\frac{(n+8) - (n-2)}{10} \right]$$

$$= \frac{1}{10} \left[n (n+2) (n+4) (n+6) (n+8) - (n-2) n (n+2) (n+4) (n+6) \right]$$

$$S = \frac{1}{10} \left[(n-1)n(n+1)(n+3)(n+5) + n(n+2)(n+4)(n+6)(n+8) + 1 \cdot 1 \cdot 3 \cdot 5 \cdot 7 \right]$$

$$T_1 = \frac{1}{10} \left[\cancel{1 \cdot 3 \cdot 5 \cdot 7 \cdot 9} - (-1) \cdot 1 \cdot 3 \cdot 5 \cdot 7 \right]$$

$$T_2 = \frac{1}{10} \left[\cancel{2 \cdot 4 \cdot 6 \cdot 8 \cdot 10} - 0 \cdot 2 \cdot 4 \cdot 6 \cdot 8 \right]$$

$$T_3 = \frac{1}{10} \left[\cancel{3 \cdot 5 \cdot 7 \cdot 9 \cdot 11} - \cancel{1 \cdot 3 \cdot 5 \cdot 7 \cdot 9} \right]$$

$$T_4 = \frac{1}{10} \left[\cancel{4 \cdot 6 \cdot 8 \cdot 10 \cdot 12} - \cancel{2 \cdot 4 \cdot 6 \cdot 8 \cdot 10} \right]$$

$$\vdots$$

$$T_{n-1} = \frac{1}{10} \left[(n-1)(n+1)(n+3)(n+5)(n+7) - \cancel{(n-3)(n-1)(n+1)(n+3)(n+5)} \right]$$

$$T_n = \frac{1}{10} \left[n(n+2)(n+4)(n+6)(n+8) - \cancel{(n-2)n(n+2)(n+4)(n+6)} \right]$$

$$S =$$

$$S = 1 \cdot 2 \cdot 3 \cdot 4 + 2 \cdot 3 \cdot 4 \cdot 5 + 3 \cdot 4 \cdot 5 \cdot 6 + \dots + (\text{upto } n \text{ terms}).$$

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